

Interest Level Management Module

Developer Documentation

Overview

The Interest Level Management Module provides functionality to mark citations and references with different levels of interest or importance within a document. This module integrates with the CKEditor and allows users to set interest levels via context menu options.

Interest Levels

The module supports three interest levels:

```
const INTEREST_LEVELS = {  
  NONE: 'none',           // Default state (no special indicator)  
  SPECIAL: 'special',     // Special importance  
  OUTSTANDING: 'outstanding' // Outstanding importance  
};
```

Core Components

1. Command Definitions

The module defines three commands corresponding to the three interest levels:

```

const commands_InterestLevel = [
  {
    name: 'INTEREST_LEVEL_NONE',
    action: 'interest_level_none',
    label: 'None',
    icon: '../UI/svg/ContextMenu/MoveBefore.svg',
    order: 210
  },
  // SPECIAL and OUTSTANDING definitions follow the same pattern
];

```

2. Key Functions

Interest Level Management

- **handleInterestLevelCommon(options, fromContext)**
Core function for handling interest level changes on DOM elements.
 - Parameters:
 - options: Object containing event, target, module context, and target finder
 - fromContext: Boolean indicating if change is from context menu
 - Returns: Array of modified target elements
- **handleInterestLevelChange_RefList(evt, target, fromContext, self)**
Handles interest level changes on reference list items.
 - Parameters:
 - evt: Event type string
 - target: Target reference list item element
 - fromContext: Boolean indicating if change is from context menu
 - self: Module context (defaults to MultiRefModule)
- **handleInterestLevelChange_Citations(evt, rid, self)**
Propagates interest level changes to all in-text citations for a reference.
 - Parameters:
 - evt: Event type string
 - rid: Reference ID
 - self: Module context

Citation Processing

- **convertTextModeToAttributeMode()**
Converts text-based interest indicators (■, ■■) to attribute-based format.

- **processCitationGroups()**
Processes citation groups to format ranges according to interest level rules.
- **restructureCitationGroup(group, ridList, interestLevels)**
Restructures a citation group based on interest levels.
 - Parameters:
 - `group`: Citation group element
 - `ridList`: Array of reference IDs
 - `interestLevels`: Array of interest levels for each reference
- **mergeAdjacentCitations(group)**
Merges adjacent citations with the same interest level.
- **createCitationElement(subRidList, level)**
Creates a new citation element.
 - Parameters:
 - `subRidList`: Array of reference IDs
 - `level`: Interest level for the citation
 - Returns: New citation DOM element

3. Editor Integration

- **setupEditorCommands_InterestLevel(editor, commands)**
Sets up editor commands and menu items for Interest Level.
 - Parameters:
 - `editor`: CKEditor instance
 - `commands`: Array of command configurations
- **initializeContextMenu(editor)**
Initializes context menu for interest level changes.
 - Parameters:
 - `editor`: CKEditor instance
- **editorListener_InterestLevel(editor)**
Editor listener to setup context menu and commands.
 - Parameters:
 - `editor`: Editor instance (optional, defaults to global editor)

Usage Examples

1. Initializing the Module

The module automatically initializes when the editor is ready:

```
document.addEventListener('DOMContentLoaded', function() {  
    CKEDITOR.on('instanceReady', function(ev) {  
        editorListener_InterestLevel(ev.editor);  
    });  
});
```

2. Changing Interest Level Programmatically

```
// Get reference element  
const referenceElement = document.querySelector('#CIT1');  
  
// Change interest level to "special"  
handleInterestLevelChange_RefList('interest_level_special', referenceElement);
```

3. Handling Annotation Preview

```
// Preview "outstanding" interest level with annotation  
handleAnnotationPreview(  
    INTEREST_LEVELS.OUTSTANDING,  
    "This is an important reference that supports our main hypothesis.",  
    MultiRefModule  
);
```

Data Model

DOM Attributes

The module uses data attributes to store interest level information:

- `data-interest-level`: Stores the current interest level
- `data-interest-level-old`: Preserves the original interest level

Context Menu Structure

The context menu for interest levels is structured as:

+ Change Interest Level

└ None

└ Special

└ Outstanding

Citation Formatting Rules

1. Single citations are displayed with their reference number and appropriate styling based on interest level.
2. Citation ranges with the same interest level are displayed as a range (e.g., "1–3") with appropriate styling.
3. Citation ranges with different interest levels are split into separate citation elements grouped by interest level.

Integration Points

- Integrates with CKEditor for context menu functionality
- Works with `MultiRefModule` for reference management
- Uses `GlobalEditor` for DOM manipulation

Error Handling

All functions include try-catch blocks with error logging via `interest_logError()` function, which logs to console and calls an external `ErrorLogTrace()` function.